The thousand grain weight experiment

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Experiment overview

- **Goal:** Investigate the effect of a growth regulator in two *oilseed* varieties (B, N)
- **Design:** Completely Randomized Design
  - Some fields received growth regulator (+)
  - Equal number of fields were used as control (-)
- **Measurement:** *Thousand Grain Weight* (TGW)
  - The weight (in grams) of a thousand seeds.
- **External effect:** Pigeons grazed some of the fields
  - Damage was varying by field
  - Measured as covariate

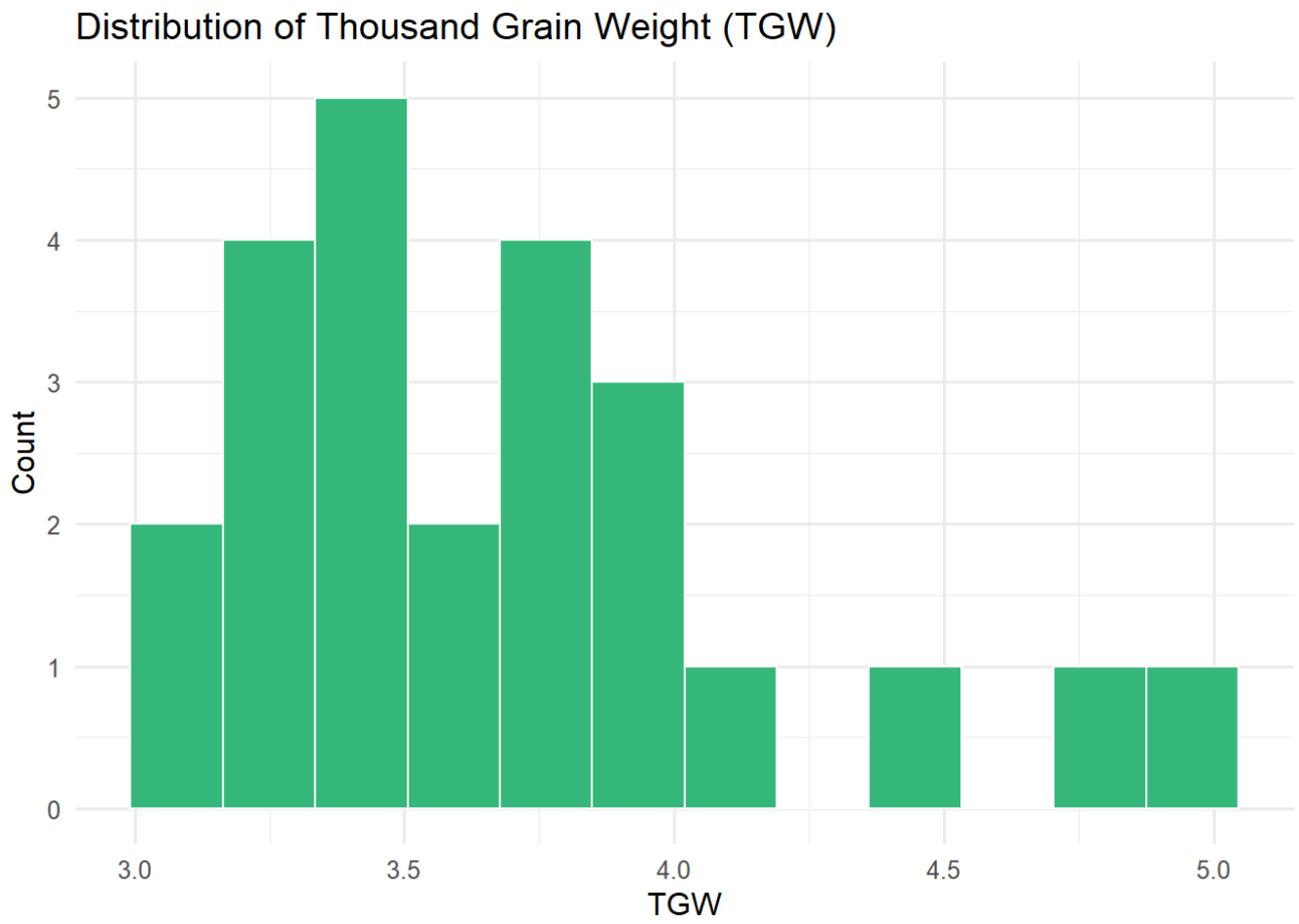
Dataset

- **Sample size: $N = 24$**
- **Variables:**
  - `Trt` (factor): Combines seed type **(B/N)** & treatment **($+/-$)**
    - Levels: $B+, B-, N+, N-$
    - 6 replicates per level
  - `TGW` (numerical): **Thousand Grain Weight** of the given field
  - `Damage` (numerical): Percentage of field damaged, rounded to the nearest **10%**

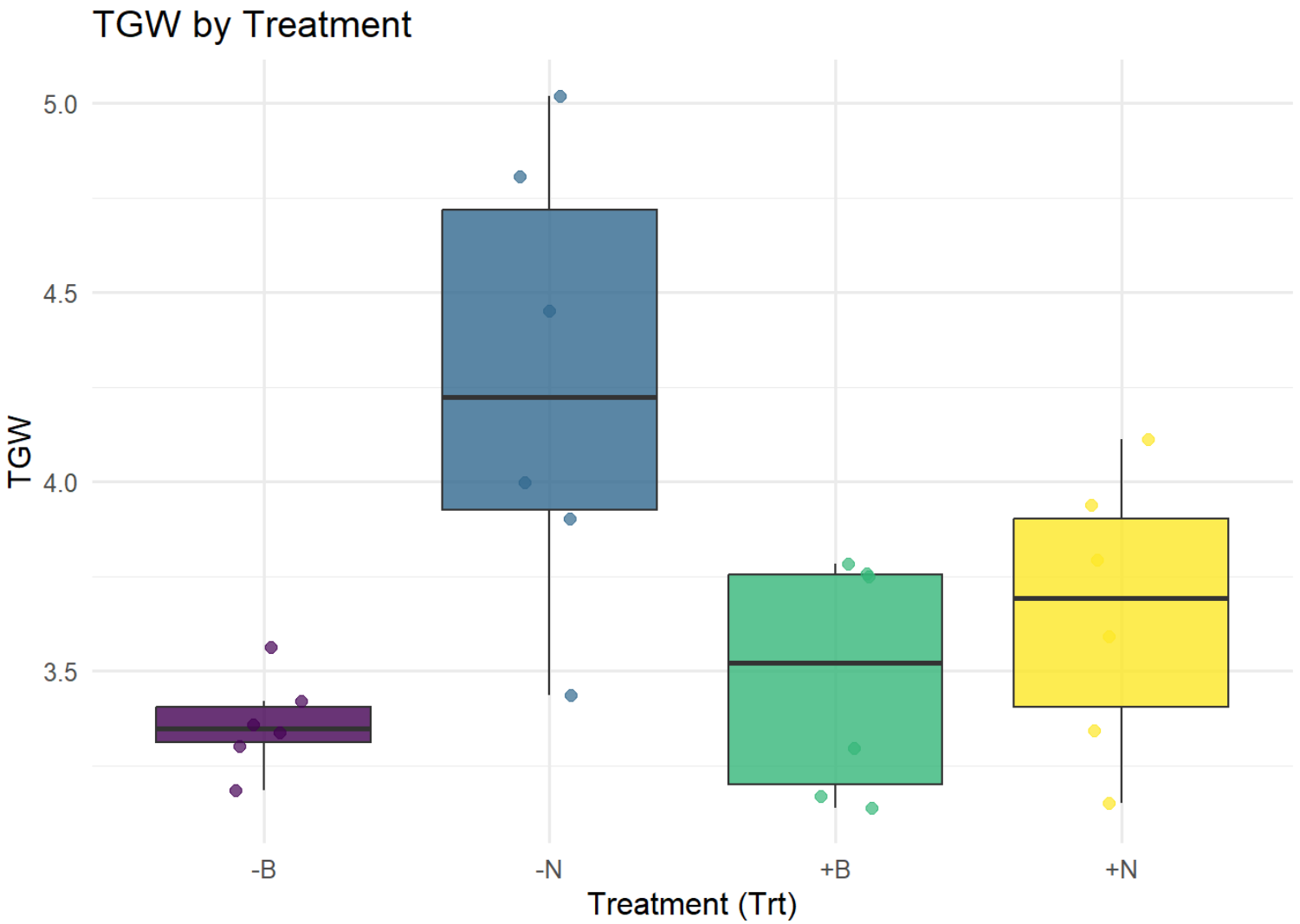
Research Question

Does `TGW` change for different levels of `Trt` while controlling for `Damage`?

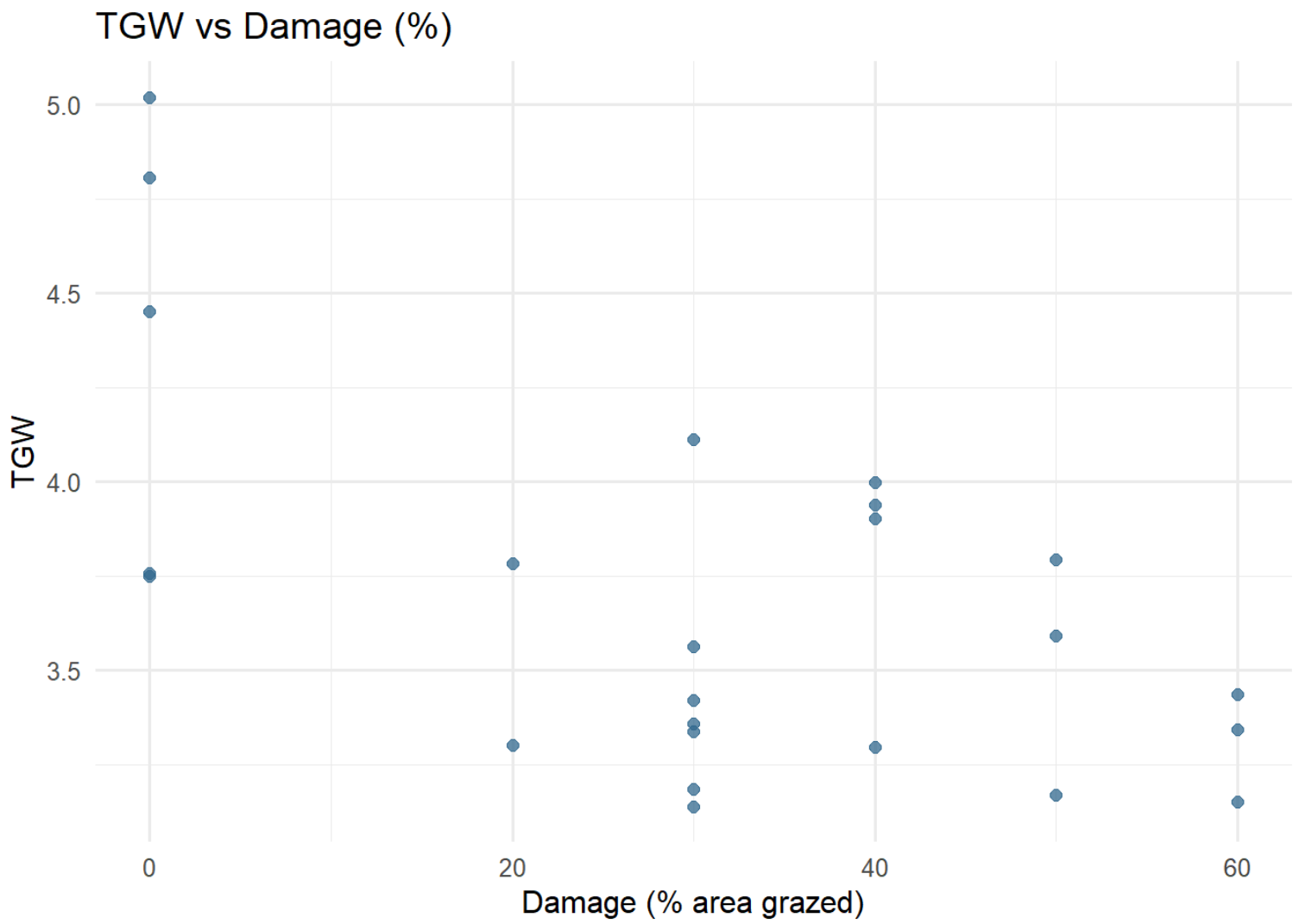
Exploratory analysis (TGW)



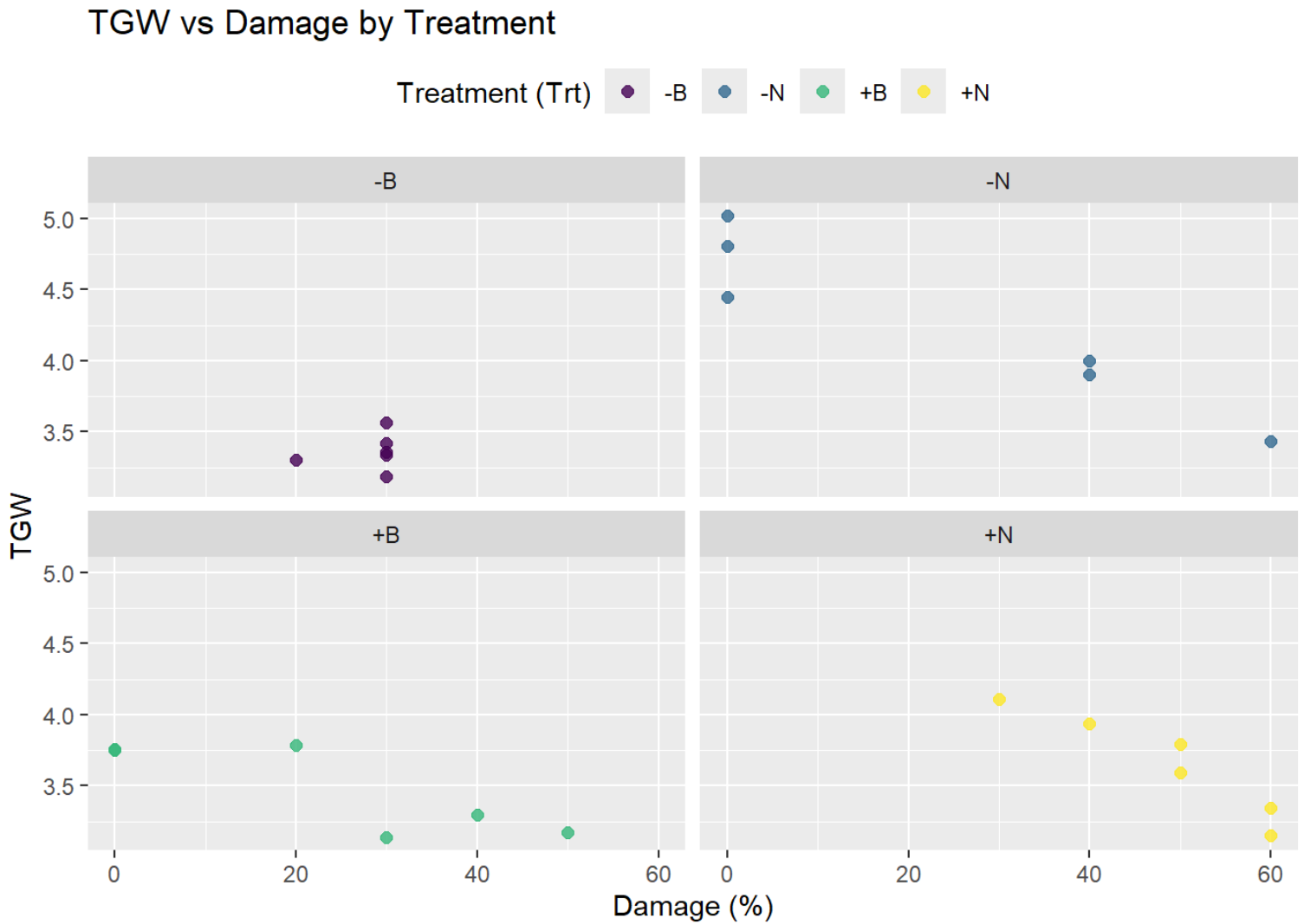
Exploratory analysis (TGW)



Exploratory analysis (Damage)



Exploratory analysis (Damage)



Models

Overview: We fit 3 models with increasing complexity.
Motivation: The true data generating process is unknown.

- **Model 1 (ANCOVA)**:
  - Minimal assumptions
- **Models 2 & 3 (Hierarchical models)**
  - Partial pooling

We then compare the three models using **LOOCV**

Model 1 (ANCOVA)

$$\text{TGW}_{ij} = \alpha + \delta_j + \beta\,\text{Damage}_{ij} + \varepsilon_{ij}, \qquad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$$

where

- α global intercept
- δ_j: effect of treatment j (fixed effect)
- β common slope for `Damage`
- σ residual SD

Model 1 - Priors

- ▷ Weakly informative priors
- ▷ Data aware
- ▷ Positive distributions for standard deviations

- $\alpha \sim \text{Normal}(\boldsymbol{m_y},\ \boldsymbol{2\, s_y})$
- $\delta_j \sim \text{Normal}(\boldsymbol{0},\ \boldsymbol{0.6\, s_y})$
- $\beta \sim \text{Normal}(\boldsymbol{0},\ \boldsymbol{3 * s_y / s_x})$
- $\sigma \sim \text{Exponential}\left(\frac{\boldsymbol{1}}{\boldsymbol{s_y}}\right)$

Model 2 (Hierarchical model, equal slopes)

$$\text{TGW}_{ij} = \alpha + \delta_j + \beta\,\text{Damage}_{ij} + \varepsilon_{ij},$$
$$\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2), \quad \delta_j \sim \mathcal{N}(0, \sigma_\delta^2)$$

where

- α: global intercept
- δ_j: effect of treatment j (random intercept)
- β: common slope for `Damage`
- σ_δ: SD of group intercepts
- σ: residual SD

Model 2 - Priors

- ▷ Weakly informative priors
- ▷ Data aware
- ▷ Positive distributions for standard deviations

- $\alpha \sim \mathcal{N}(m_y, 3\,s_y)$
- $\beta \sim \mathcal{N}\left(0, 3\,\frac{s_y}{s_x}\right)$
- $\sigma \sim \text{Exponential}\left(\frac{1}{s_y}\right)$
- $\sigma_\delta \sim \mathcal{N}^+(0, 3\,s_y)$

Model 3 (Hierarchical with unequal slopes)

$$\text{TGW}_{ij} = \alpha + \delta_j + (\beta + \gamma_j)\,\text{Damage}_{ij} + \varepsilon_{ij},$$
$$\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2), \delta_j \sim \mathcal{N}(0, \sigma_\delta^2), \gamma_j \sim \mathcal{N}(0, \sigma_\gamma^2)$$

- α: global intercept
- δ_j: effect of treatment j (random intercept)
- β: global slope for Damage
- γ_j: effect of treatment j (random slope)
- σ_δ: SD of group intercepts
- σ_γ: SD of group slopes
- σ: residual SD

Model 3 - Priors

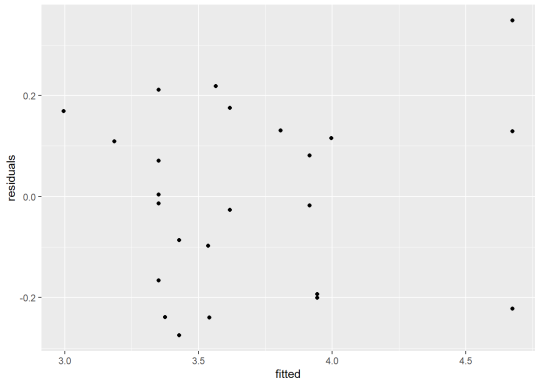
- ▷ Weakly informative priors
- ▷ Data aware
- ▷ Positive distributions for standard deviations

- $\alpha \sim \mathcal{N}(m_y, 3\,s_y)$
- $\beta \sim \mathcal{N}\left(0, 3\,\frac{s_y}{s_x}\right)$
- $\sigma \sim \text{Exponential}\left(\frac{1}{s_y}\right)$
- $\sigma_\delta \sim \mathcal{N}^+(0, 3\,s_y)$
- $\sigma_\gamma \sim \mathcal{N}^+\left(0, 3\,\frac{s_y}{s_x}\right)$

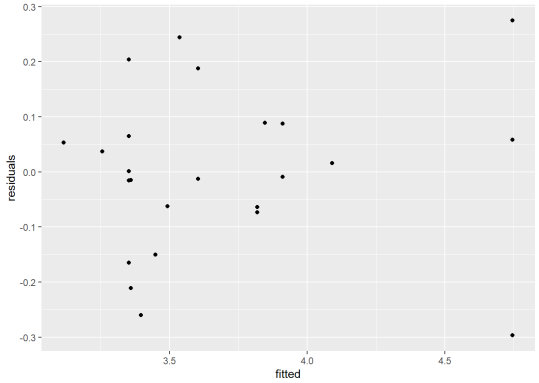
Methods

- All 3 models fit with No-U-Turn Sampler (NUTS)
- 2000 total iterations per chain
- 1000 warmup iterations
- 4 chains
- Model-specific parameters:
  - Model 1: `adapt_delta` $= 0.9$
  - Model 2: `adapt_delta` $= 0.995$, `max_tree_depth` $= 15$
  - Model 3: `adapt_delta` $= 0.99$, `max_tree_depth` $= 15$

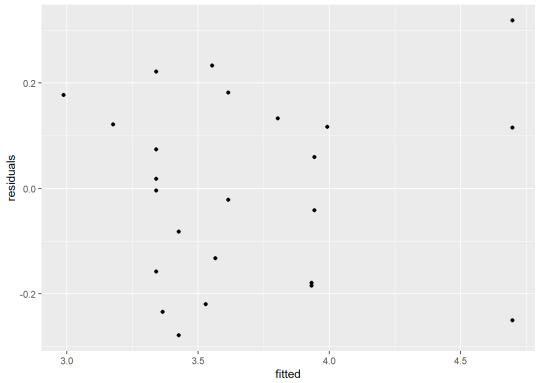
Model Diagnostics



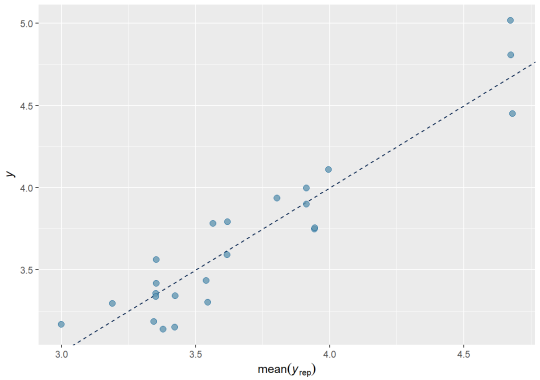
Model 1 Residuals



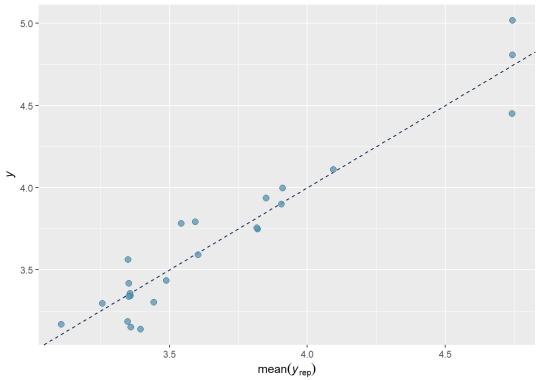
Model 2 Residuals



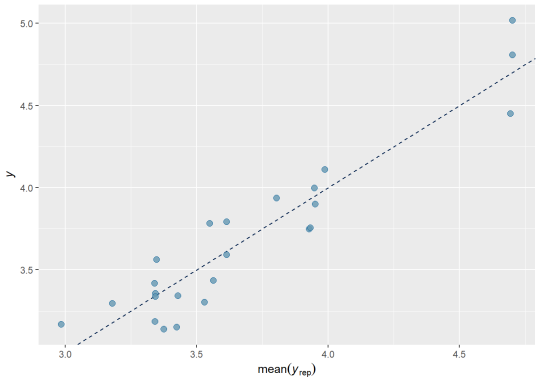
Model 3 Residuals



Model 1 PPC



Model 2 PPC



Model 3 PPC

Model Comparison

- All models have good fits $\rightarrow$ we can compare them
- We use Leave-One-Out Cross-Validation (LOOCV)
- We also include a degenerate model (without covariates or treatments) to assess overall effect

Model	**elpd_diff**	**se_diff**
Model 3	0.0	0.0
Model 2	-1.0	1.7
Model 1	-1.0	1.9
No effect model	-22.5	4.2

- Similar performance between the 3 models
- Choose Model 1 (simplest)

Model summary

- Model summary: (-B reference group)

Parameter	Estimate	l-95% CI	u-95% CI	Rhat	Bulk ESS	Tail ESS
α	3.69	3.61	3.78	**1.00**	**3363**	**2452**
δ_1	-0.36	-0.49	-0.21	**1.00**	**3581**	**2676**
δ_2	0.40	0.24	0.54	**1.00**	**2763**	**2729**
δ_3	-0.33	-0.47	-0.19	**1.00**	**2961**	**2764**
β	-0.02	-0.02	-0.01	**1.00**	**4735**	**3015**
σ	0.20	0.15	0.29	**1.00**	**2539**	**2444**

- β has significant effect
- Did the treatments have significant differences?

Comparison of treatment effects

- We use Estimated Marginal Means to compare the effect the regulator had on the different seeds.

Contrast	Estimate	Lower HPD	Upper HPD
$(-B) - (-N)$	-0.7611	-0.978	-0.534
$(-B) - (+B)$	-0.0255	-0.247	0.222
$(-B) - (+N)$	-0.6487	-0.898	-0.405
$(-N) - (+B)$	0.7371	0.503	0.958
$(-N) - (+N)$	0.1126	-0.149	0.378
$(+B) - (+N)$	-0.6222	-0.872	-0.352

Conclusion:

- Growth regulator: **No effect**
- Type of seed responsible for differences
- Damage had a small but significant effect

Limitations

- Results not conclusive
- Small dataset
- Damage was rounded
- More covariates relevant